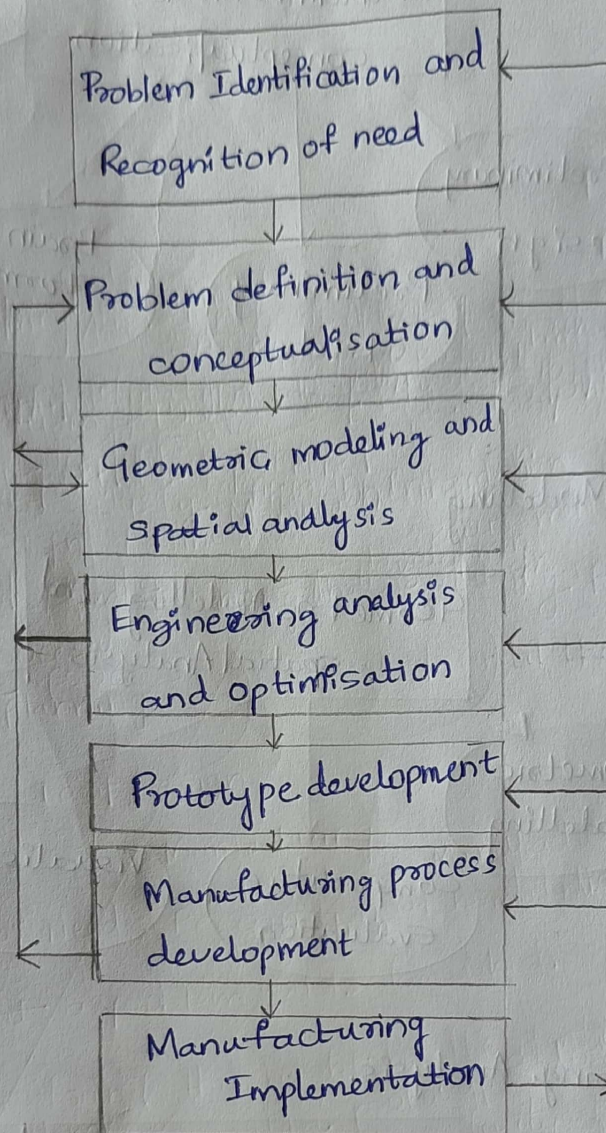
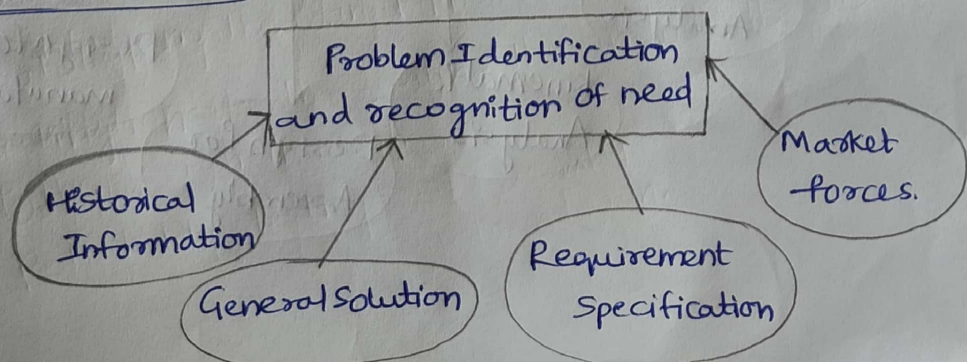


- 1) Briefly Explain the Conventional process of the product cycle in conventional manufacturing environment.

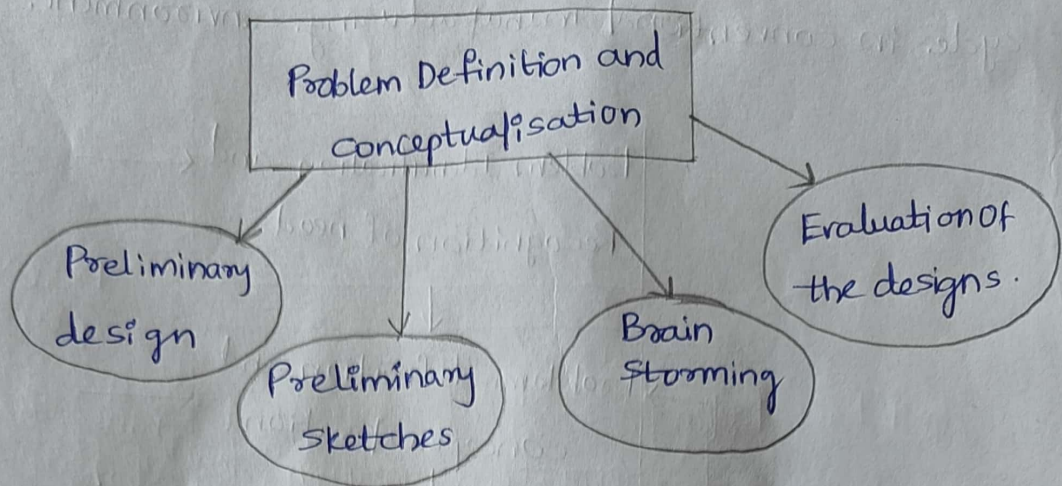


stages in the design Process

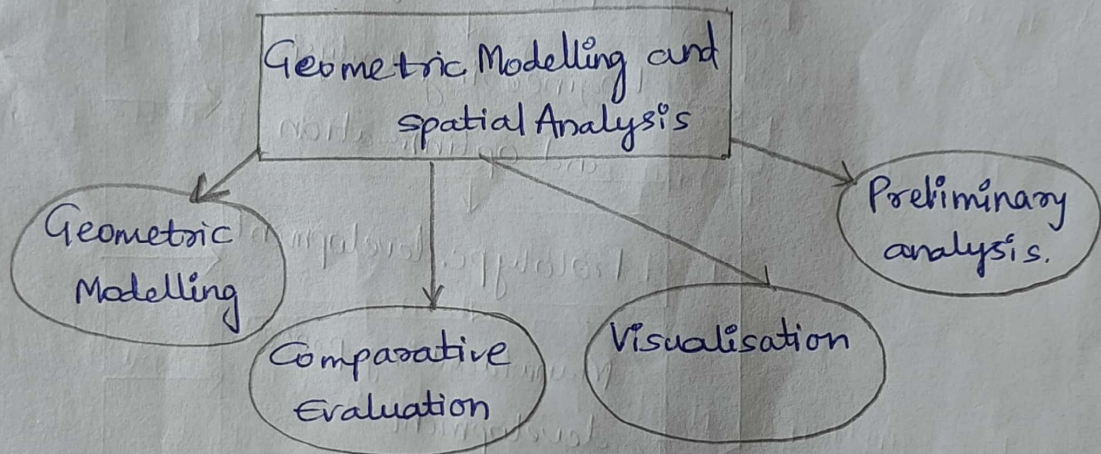
1. Problem Identification:



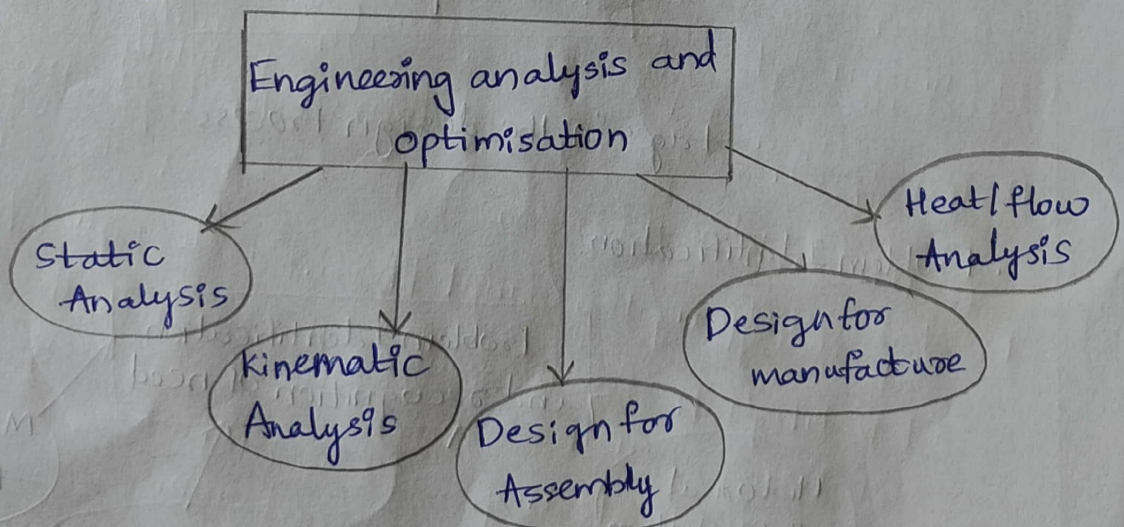
2. Problem Definition



3. Geometric Modelling



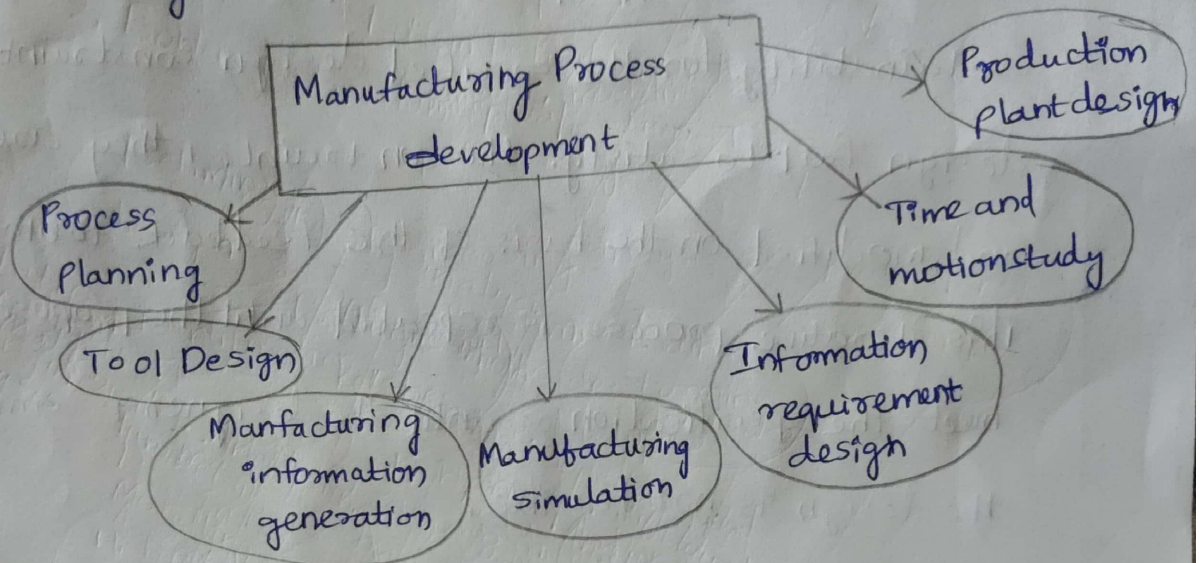
4. Engineering Analysis.



5. Prototype Development

1. Rapid Prototype
2. Test and Evaluation
3. Design Refinement
4. working drawings

6. Manufacturing Process Development



A. Recognition of Need: Recognition of need involves the realization by someone that a problem exists for which some feasible solution is to be found. This might be the identification of some defect in a current machine design activity by an engineer or the perception of a new product marketing opportunity by salesman.

B. Definition of problem: Definition of problem involves through specification of the item to be designed. This specification will generally include functional and physical characteristics, cost, quality, performance, etc.

C. Synthesis: During the synthesis phase of the design process various preliminary ideas are developed through research of similar product or design in use.

D. Analysis and optimization: The resulting preliminary design is then subjected to appropriate analysis to determine their sustainability for the specified design constraint. If this design fail to satisfy the constraints, they are redesigned and modified on the basis of the feedback from the analysis. This iterative process is repeated until the proposed design meets the specification or until the designer is convinced that the design is not feasible.

E. Evaluation: The Assessment or evaluation of the design against the specification established during the problem definition phase is then carried out. This often requires the fabrication and testing of a prototype model to evaluate operating performance quality, reliability etc.

- Prototyping
- Testing
- Quality
- Reliability Testing

F. Presentation: The final Phase of design process is the presentation of the design. This include the documentation of the design through drawing, material specification, assembly

2. what are the input devices & o/p devices. Explain them briefly.

Input devices:

- * No. of input devices is available. These devices are used to input two possible types of information text and graphics.

- * Text - input devices and the alphanumeric keyboards.

- * There are two classes of graphics input devices: Locating devices and image input devices.

1. Scanners: Scanners comprise other class of graphics -input device

There are four relevant parameters to measure the performance of graphics input devices. These are resolution, accuracy, repeatability, and linearity.

2. Keyboards: Keyboards are typically employed to create / edit programs or to perform word processing functions.

3. Digitizing Tablets: It is locating and pointing device.

It is a small, low-resolution digitizing board often used in conjunction with a graphics display.

- * It is a flat surface over which a stylus can be moved by the user.

- * A tablet's typical resolution is 200 dots per inch.

Mouse: There are two basic types of mouse available mechanical and optical.

* The mechanical mouse has roller in order to record the mouse motion in x and y directions.

Joysticks and Trackballs:

* The joystick works by pushing its stick backwards or forward or to left or right. The extreme positions of these directions correspond to the four corners of the screen.

* Trackball is similar in principal to a joystick but allows more precise fingertip control.

Output devices:

1. Printer: Its primary function is to transform a text document or image we can see on the screen to a physical format on paper.

Monitor: It is used to view data and images from computer.

Projector: It works by projecting a presentation of images or videos on a projection screen. The information is received directly from the computer, and the projector acts as a transmitter.

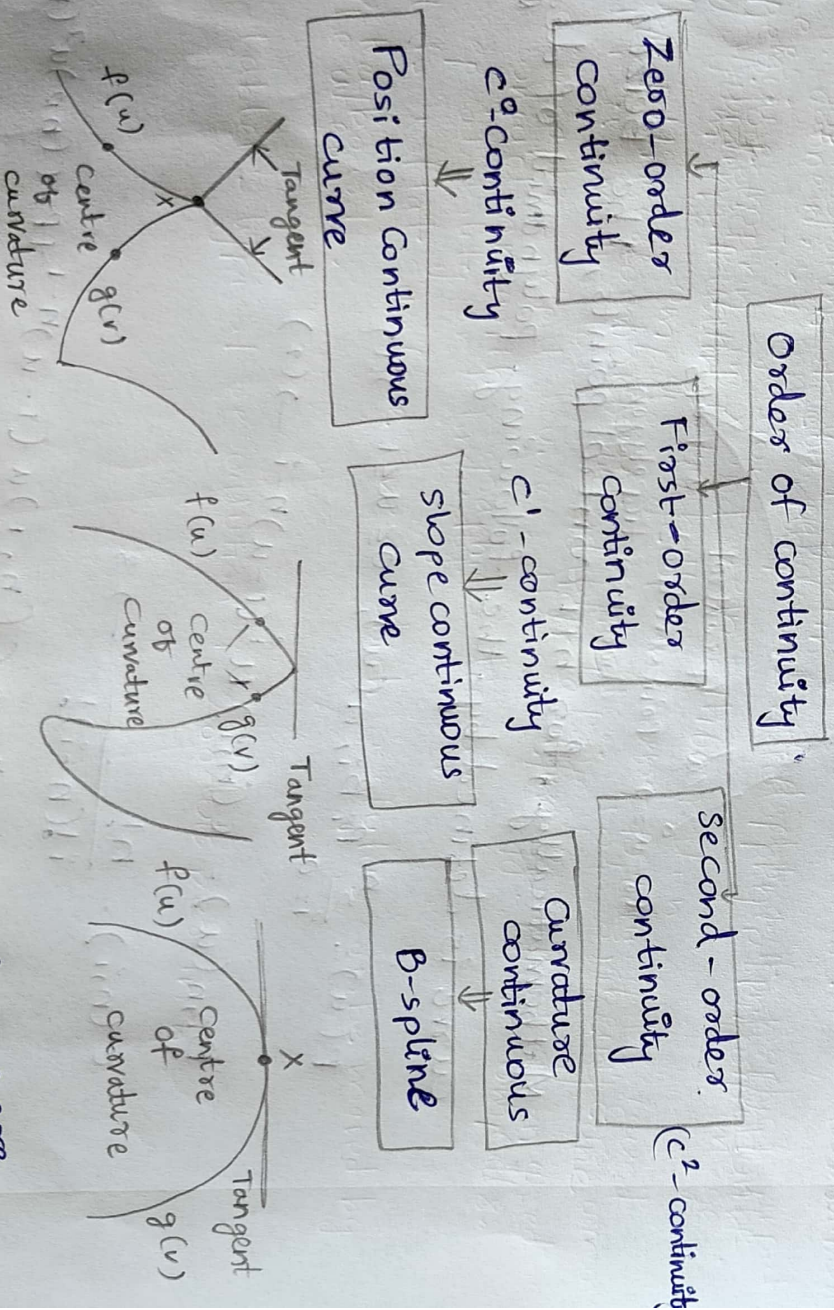
Plotters: For printing large plans or projects, being one of the printers most used by designers and architects.

Sound card: It controls the output of sound signals, allowing devices such as speakers and headphones to function.

What is meant by continuity of curves. What are types of continuity curves. Explain them in brief.

Curvature Continuity: The order of continuity becomes important when a complex curve is modeled by several curve segments pieced together end to end.

The Continuity of Surface refers to the smoothness and quality of the shape.



Two boundary curve segments shown in above figure are meeting at a vertex v . Two curves are $f(u)$ and $g(v)$ in intervals $[a, b]$ and $[m, n]$ respectively.

Consider endpoint $f(u)$ and start point $g(m)$.

If $f(b)$ and $g(m)$ are equal then curves are continuous

at $f(b)=g(m)$

if for all $i \leq k$, $f(b)=g(m)$ then

c^k continuity at point $f(b)=g(m)$. Intuitively,

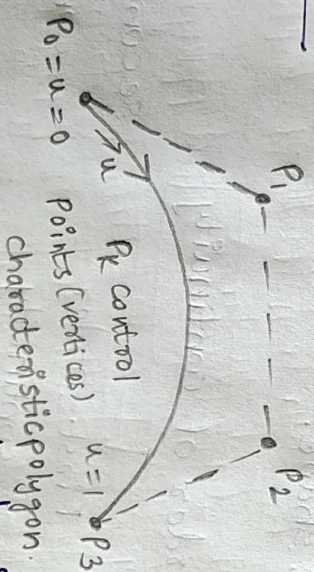
c^0 - Continuity of end point only (or) continuity of position.

c^1 - Tangent continuity (or) first derivative if position

c^2 - Hydrodynamic character, light reflection

Curvature continuity (or) second derivative of position

Bezier Curve:



Mathematically, for $n+1$ control points, the Bezier curve is defined by the following polynomial of degree n .

$$P(u) = \sum_{i=0}^n P_i B_{i,n}(u), \quad 0 \leq u \leq 1 \quad \rightarrow (1)$$

$$B_{i,n}(u) = C(n,i) u^i (1-u)^{n-i} \rightarrow (2)$$

$$C(n,i) = \frac{n!}{i!(n-i)!} \rightarrow 3$$

$$P(u) = P_0(1-u)^n + P_1 C(n,1)u(1-u)^{n-1} + P_2 C(n,2)u^2(1-u)^{n-2} + \dots + P_{n-1} C(n,n-1)u^{n-1}(1-u) + P_n u^n, \quad 0 \leq u \leq 1 \rightarrow (4)$$